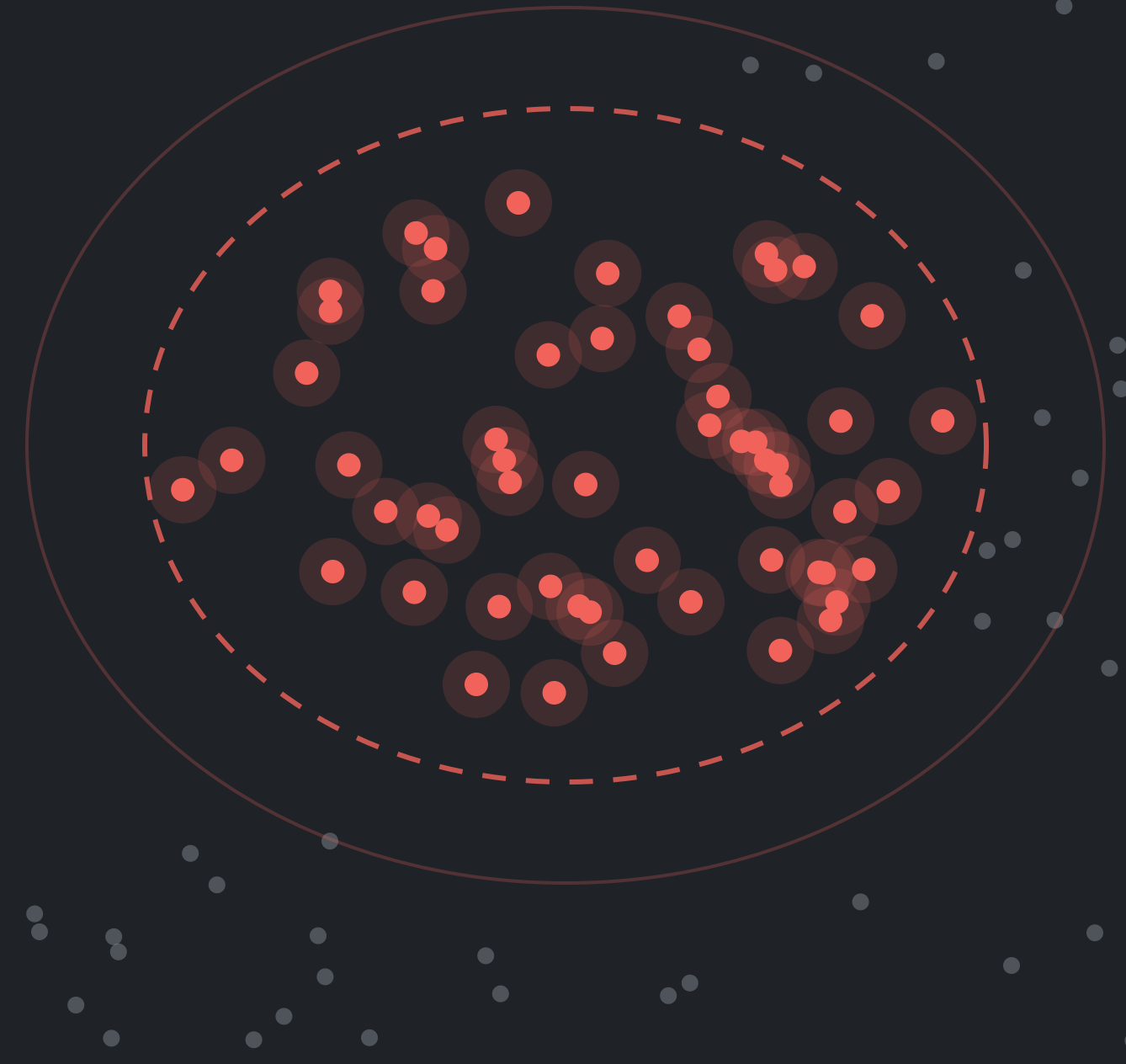


Inbox Atlas

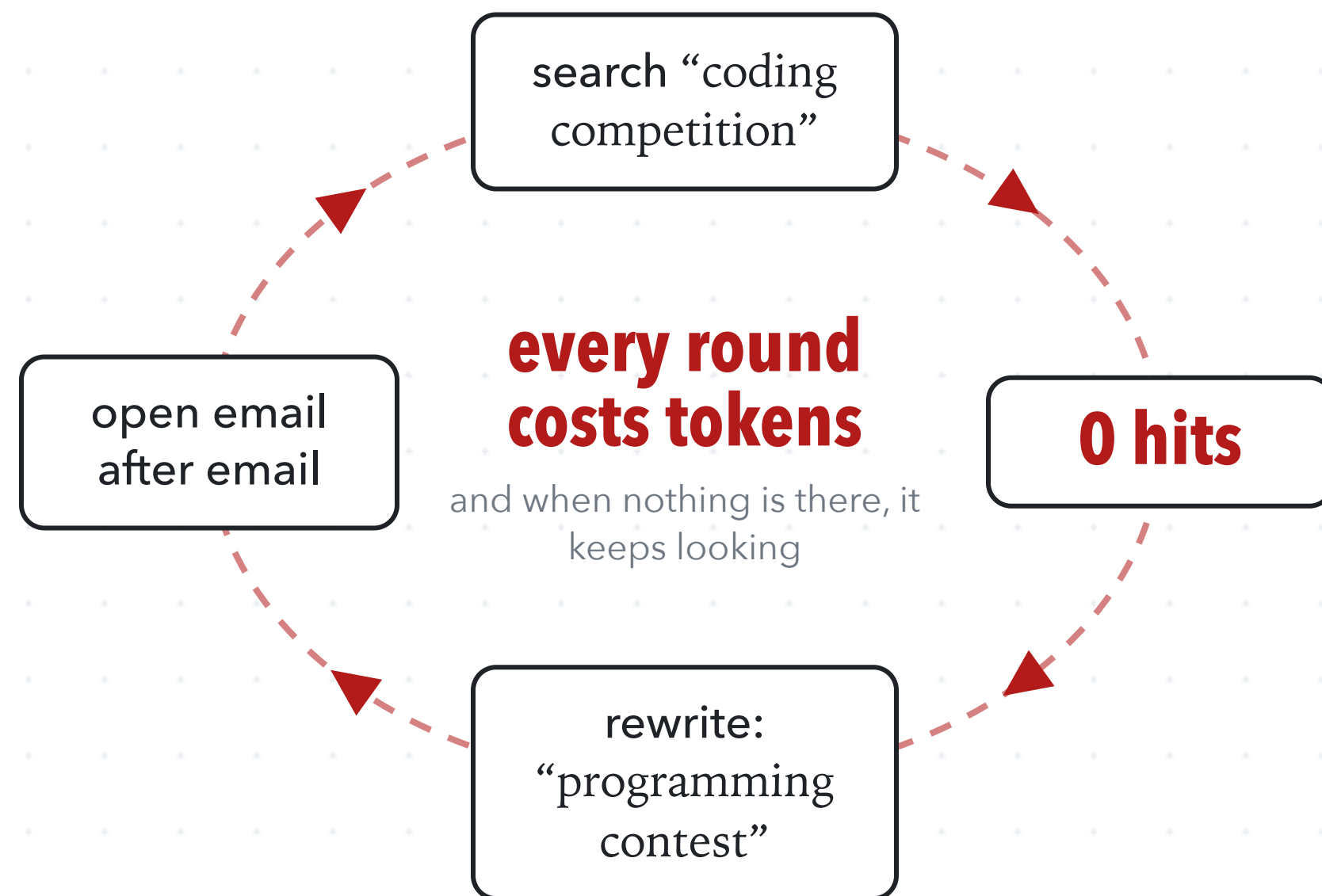
Your agents read 3,700 tokens to answer one email question. Inbox Atlas hands them the 225 that matter, or tells them nothing is there.

Avinash Senthil, Antranig Baghdassarian, Chelsea Lin, Emily Han

BigRed//Hacks 2026, Navigation



Agents search your mail by matching words.



The words the model writes are often not in the email.

the agent searches

“coding competition”

the email says

Codeforces Round 1043 (Div. 2) starts in 24 hours

missed

keyword search returns

Online assessment for Software Engineering Intern

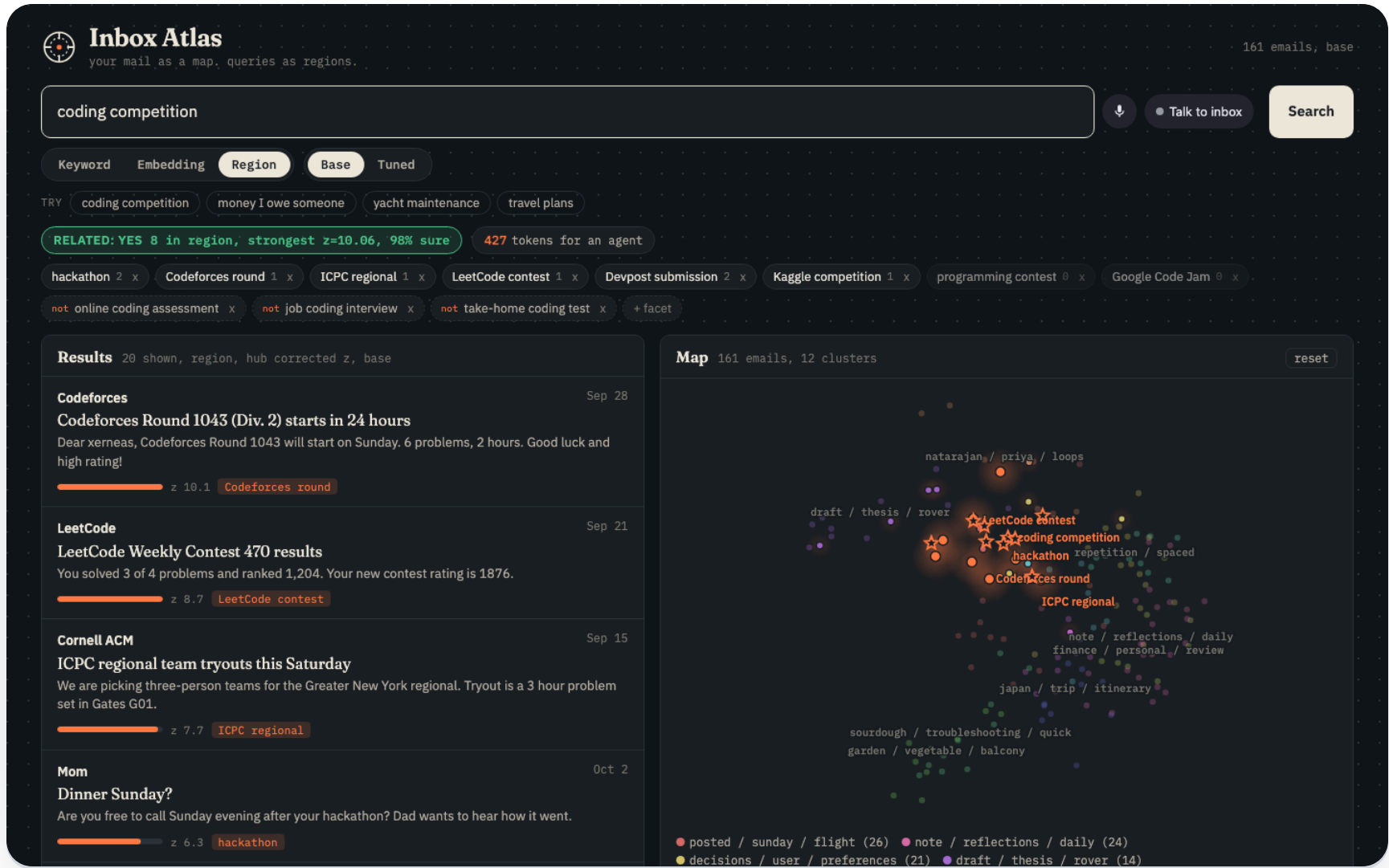
a job test

7 of 12

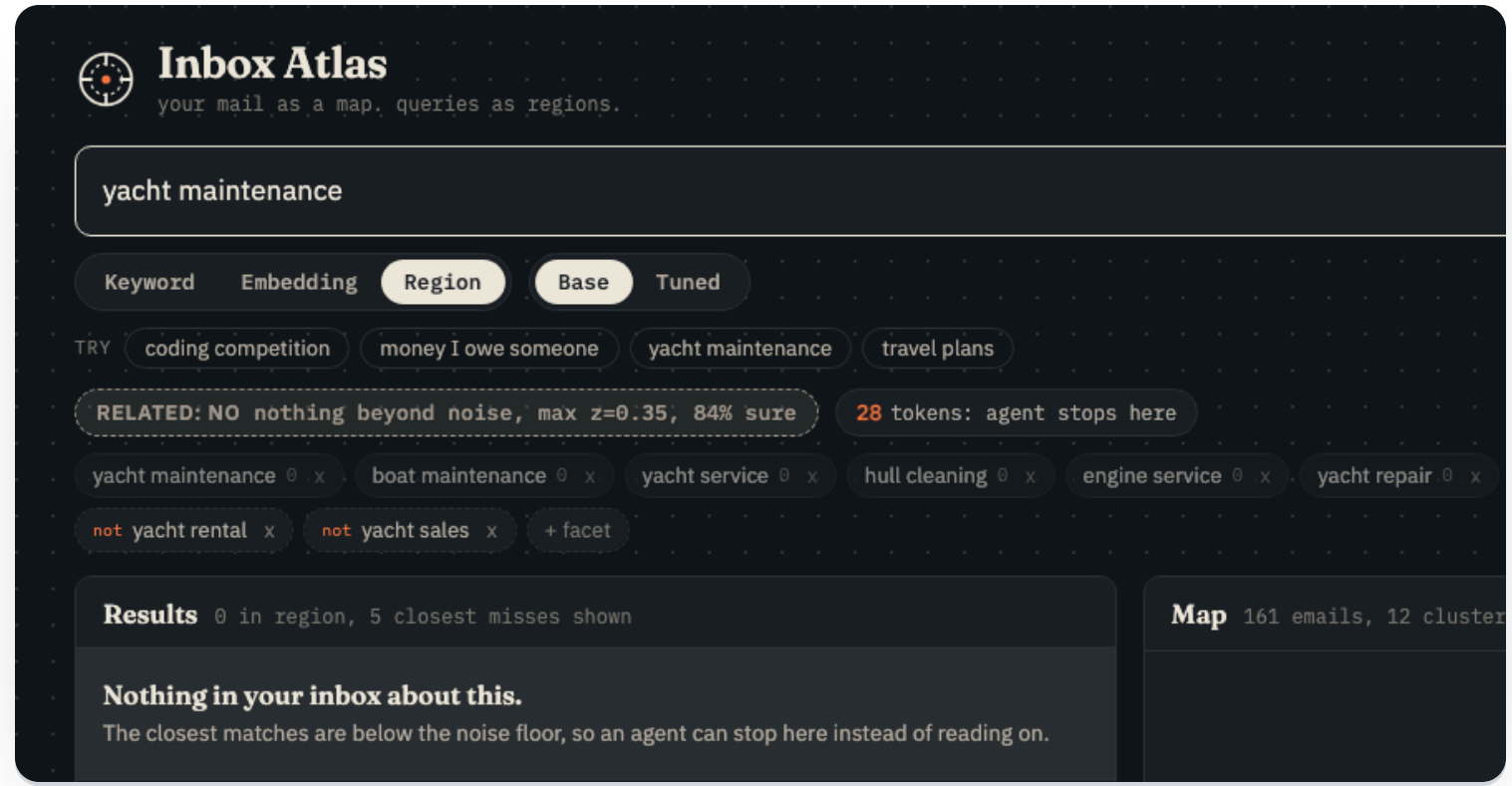
meaning-based questions
where keyword search found
nothing at all

So we built Inbox Atlas.

A retrieval layer that searches by meaning and returns a decision, for agents and people.



Search by meaning: Grok's facets and exclusions, **RELATED: YES**, the region glowing on the map.



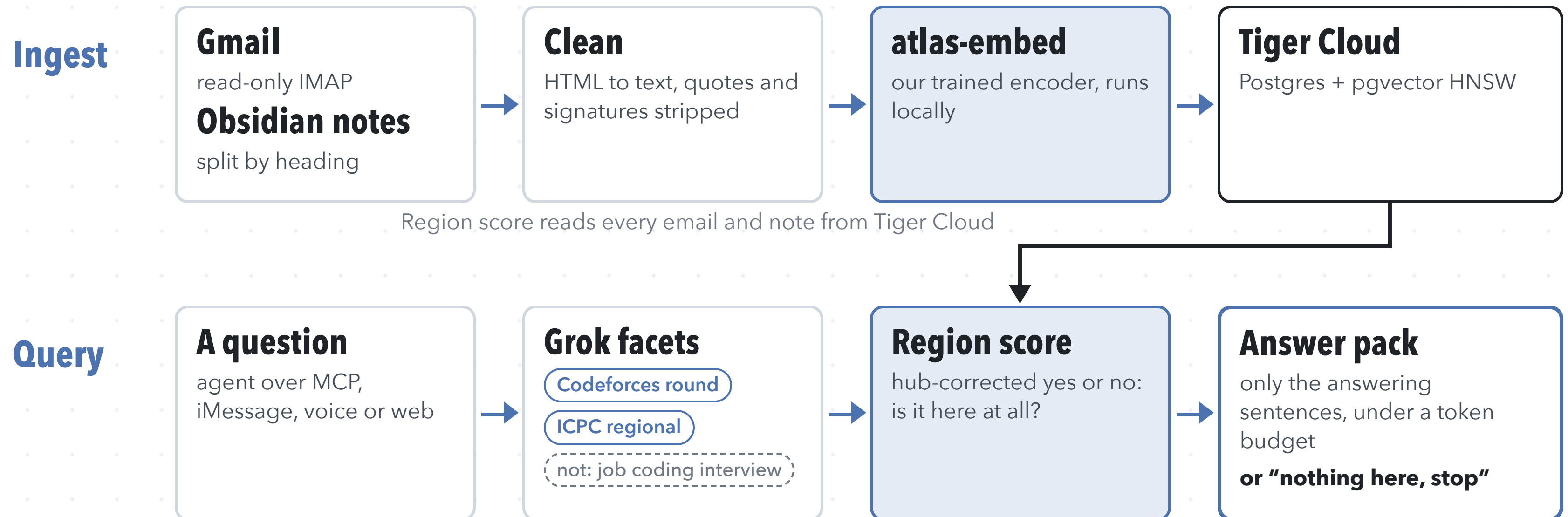
Not there: **RELATED: NO**, 28 tokens, the agent stops.

did anyone ask me for money?

Yes. Venmo Wed Sep 30: Alex requested \$42.50 for pizza + uber friday.

watch internships

The whole pipeline, end to end.



17x fewer tokens on email questions, and more accurate.

Tokens per question	keyword	3,724
	Atlas	225

Answer accuracy	keyword	33%
	Atlas	58%

Finding the right emails Recall@10	keyword	0.175
	Atlas	0.867

12 demo inbox questions, answered and graded by Grok from only the context each method hands over.

Why it is more accurate

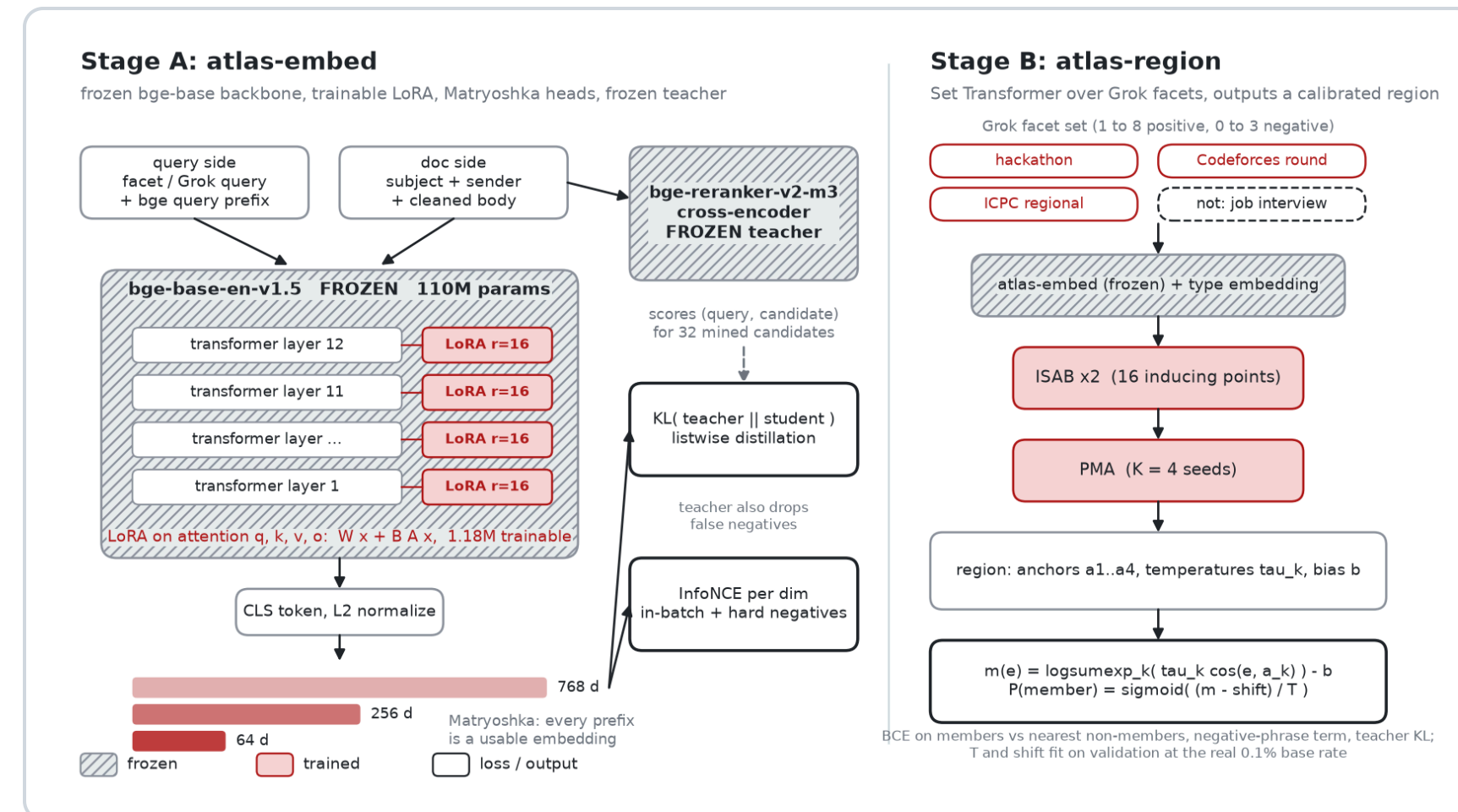
The words a model would search for are often not in the email. Word matching misses them. Meaning does not.

“coding competition”

finds

“Codeforces Round 1043”

How we built it.



1.18M

LoRA params, frozen bge-base

491k

frozen teacher scores

224,867

Enron emails

1 GPU

RTX 4070 SUPER, overnight

19,980

Grok labels

0.692 vs 0.635

nDCG@10, vague queries

Grok

Facets and exclusions, the tool-calling agent, speech to text, realtime voice.

Tiger Cloud

Postgres with pgvector HNSW, a TimescaleDB query log, chat history.

Photon

The two-way iMessage agent on Spectrum.

MCP

One server for Cursor, Claude Code, Hermes and OpenClaw.

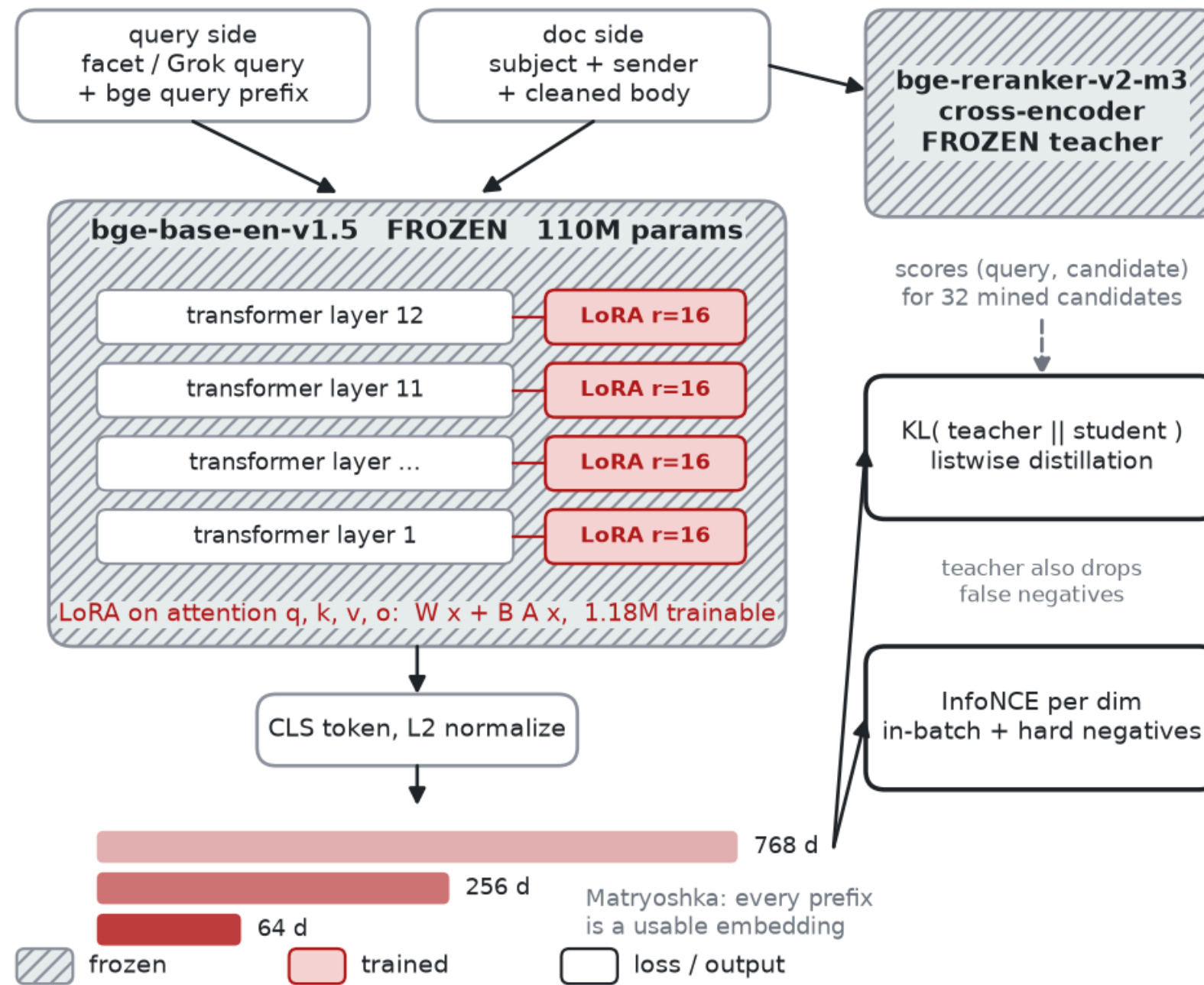
Cursor CLI

Built and routed with it. In our own testing outside this repo, Cursor CLI was the most token-efficient at routing compared with Claude Code, Codex and Grok.

Inside the model.

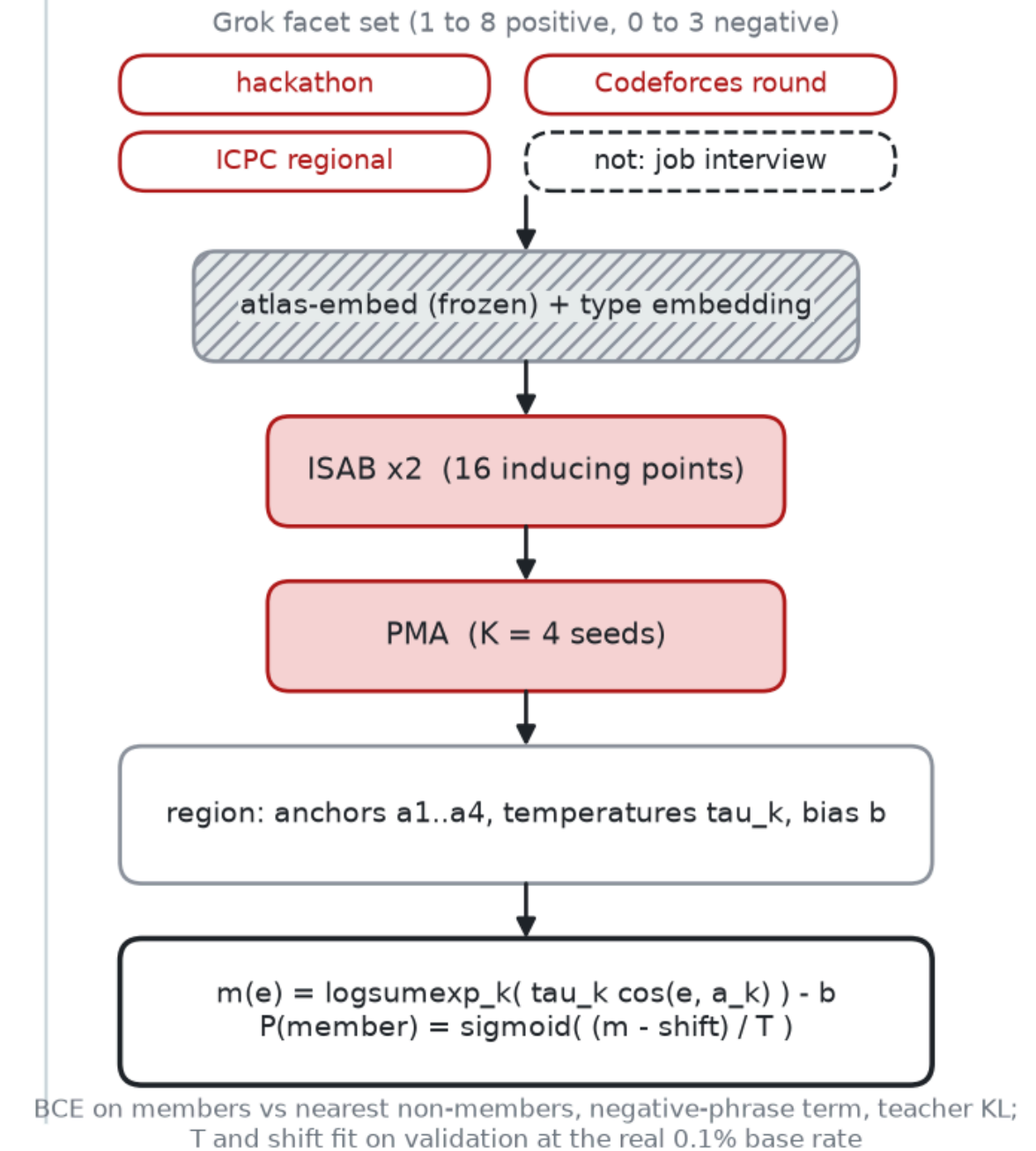
Stage A: atlas-embed

frozen bge-base backbone, trainable LoRA, Matryoshka heads, frozen teacher



Stage B: atlas-region

Set Transformer over Grok facets, outputs a calibrated region



Keyword search vs Inbox Atlas.

	keyword	Atlas	gain
Questions with zero relevant hits 12 meaning-based questions	7 of 12	0 of 12	7 to 0
Recall@10 finding the right emails	0.175	0.867	about 5x
Tokens per email question	3,724	225	17x fewer
Answer accuracy	33%	58%	more accurate
Tokens when the topic is absent	1,858	31	60x fewer
Real 419 note vault tokens handed to the agent	10,907	625	17x fewer

RAG always returns k chunks. We return a decision.

A building block for every second brain.

Obsidian vaults

agent memory folders

one MCP server for

Hermes

OpenClaw

Cursor

Claude Code

"what pulse widths did I use for the arm servos?"

📁 projects/

📁 robotics/

📁 **arm/**

📄 **Servo calibration.md#pulse-widths**

Elbow servo: 610 to 2380 microseconds.

210 tokens

Folder first, then the note, then the exact line.

23 tokens

"did I book a yacht charter?" Nothing here. It just stops.

Live demo

Search by meaning in the browser



Text it on iMessage



Ask the agent in Cursor

Impact and what's next.

Today

17x

fewer tokens on email questions:
3,724 to 225

33% to 58%

answer accuracy, at those 17x fewer
tokens

31

tokens when the topic isn't there, so
agents stop

625

tokens instead of 10,907 on a real
419 note vault

Next

Fine-tune on each person's own mail

Calendar, docs and chat logs behind the same tool

Agent feedback turned into training pairs

Any second brain or agent memory, one MCP server

Navigation used to mean maps. **Now it also means navigating what we and our agents know.**

github.com/xerneas3318/inbox-atlas

Avinash Senthil, Antranig Baghdassarian, Chelsea Lin, Emily Han